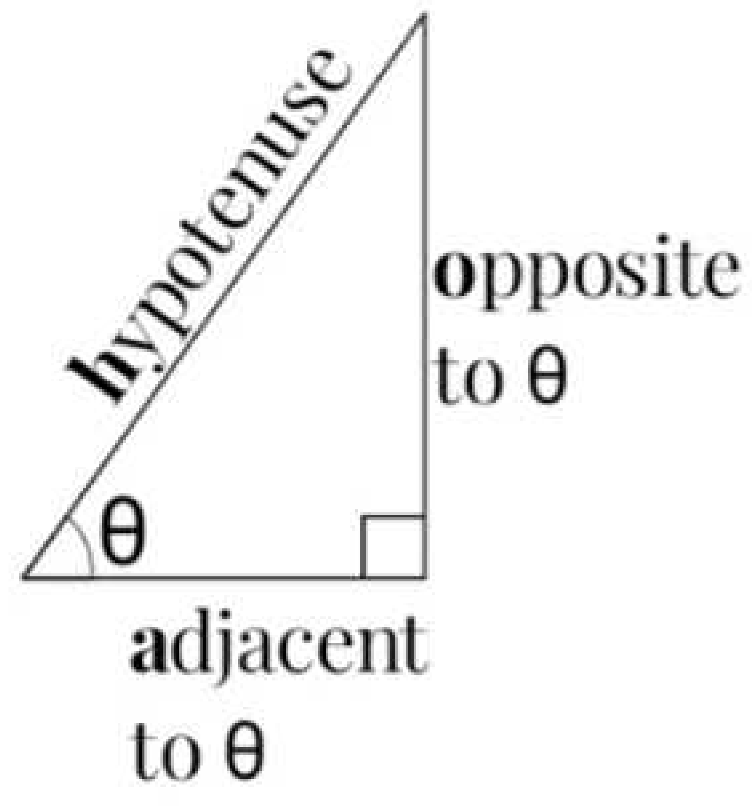
Trig Identities

- A quick recap of trigonometry :



SOH CAH TOA :

$\sin\theta = \frac{o}{h}$

$\cos\theta = \frac{a}{h}$

$\tan\theta = \frac{o}{a}$

- Let's derive 2 important identies :

(1) $\tan\theta = \frac{o}{a}$ But $\sin\theta = \frac{o}{h}$ gives $o = h\sin\theta$

And $\cos\theta = \frac{a}{h}$ gives $a = h\cos\theta$

$\therefore \tan\theta = \frac{h\sin\theta}{h\cos\theta}$

$$\boxed{\tan\theta = \frac{\sin\theta}{\cos\theta}}$$

(2) The pythagorian identity :

$h^2 = o^2 + a^2$

$h^2 = h^2\sin^2\theta + h^2\cos^2\theta$

$\frac{\cancel{h^2}}{\cancel{h^2}} = \frac{\cancel{h^2}\sin^2\theta}{\cancel{h^2}} + \frac{\cancel{h^2}\cos^2\theta}{\cancel{h^2}}$ Divide each term by h^2

$\boxed{\therefore 1 = \sin^2\theta + \cos^2\theta}$ *Note* : $\cos^2\theta = \cos\theta.\cos\theta$

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Other important things to know :

(1) $\cos^{-1}\theta \neq \dfrac{1}{\cos\theta}$

(2)(i) $\dfrac{1}{\cos\theta} = \sec\theta$

(ii) $\dfrac{1}{\sin\theta} = \text{cosec}\theta$

(iii) $\dfrac{1}{\tan\theta} = \cot\theta \quad \left[\dfrac{1}{\left(\frac{\sin\theta}{\cos\theta}\right)} = \dfrac{\cos\theta}{\sin\theta}\right]$

COMPOUND ANGLE IDENTITIES :

same sign

(1) $\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B$

$\sin(A+B) = \sin A\cos B + \cos A\sin B$

$\sin(A-B) = \sin A\cos B - \cos A\sin B$

opposite signs

(2) $\cos(A \pm B) = \cos A\cos B \pm \sin A\sin B$

$\cos(A+B) = \cos A\cos B - \sin A\sin B$

$\cos(A-B) = \cos A\cos B + \sin A\sin B$

same sign

(3) $\tan(A \pm B) = \dfrac{\tan A \pm \tan B}{1 \pm \tan A\tan B}$

opposite signs

$\tan(A+B) = \dfrac{\tan A + \tan B}{1 - \tan A\tan B}$

$\tan(A-B) = \dfrac{\tan A - \tan B}{1 + \tan A\tan B}$

Therefore for double angles :

(1) $\sin 2A = \sin(A+A)$

$= \sin A\cos A + \cos A\sin A$

$\boxed{\sin 2A = 2\sin A\cos A}$

(2) $\cos 2A = \cos(A+A)$

$= \cos A\cos A - \sin A\sin A$

$$\boxed{\cos 2A = \cos^2 A - \sin^2 A}$$

(3) $\tan 2A = \tan(A+A)$

$$= \frac{\tan A + \tan A}{1 - \tan A \tan A}$$

$$\boxed{\tan 2A = \frac{2\tan A}{1 - \tan^2 A}}$$

Degrees($^\circ$) and radians(^{c})

- Both are the units for measurement of angle and you must also be familar with their conversion.

- $180^\circ = \pi^c$ OR $360^\circ = 2\pi^c$

- $x^\circ = \frac{\theta}{\pi} \times 180^\circ$ OR $x^c = \frac{\theta}{180} \times \pi^c$

Examples

1) Find an equivalence of each of the following angles in

(a) degrees

(i) $\frac{\pi}{2}^c$

(ii) $\frac{3\pi}{2}^c$

(b) radians

(i) 720°

(ii) 15°

sol:

1(a)(i) $x^\circ = \frac{\theta}{\pi} \times 180^\circ$

$$= \frac{\left(\frac{\pi}{2}\right)}{\pi} \times 180^\circ$$

$$= \frac{\pi}{2} \times \frac{1}{\pi} \times 180^\circ = 90^\circ$$

(ii) $x^\circ = \frac{\theta}{\pi} \times 180^\circ$

$$= \frac{\left(\frac{3\pi}{2}\right)}{\pi} \times 180^\circ$$

$$= \frac{3\pi}{2} \times \frac{1}{\pi} \times 180^\circ = 270^\circ$$

b(i) $x = \frac{\theta}{180} \times \pi^c$

$= \frac{720}{180} \times \pi^c$

$= 4\pi^c$

(ii) $x = \frac{\theta}{180} \times \pi^c$

$= \frac{15}{180} \times \pi^c$

$= \frac{1}{12}\pi^c$

<u>Practice</u>

1) Convert the following

(a) degrees to radians

(i) 285°

(ii) 330°

(iii) 45°

(b) radians to degrees

(i) $4\pi^c$

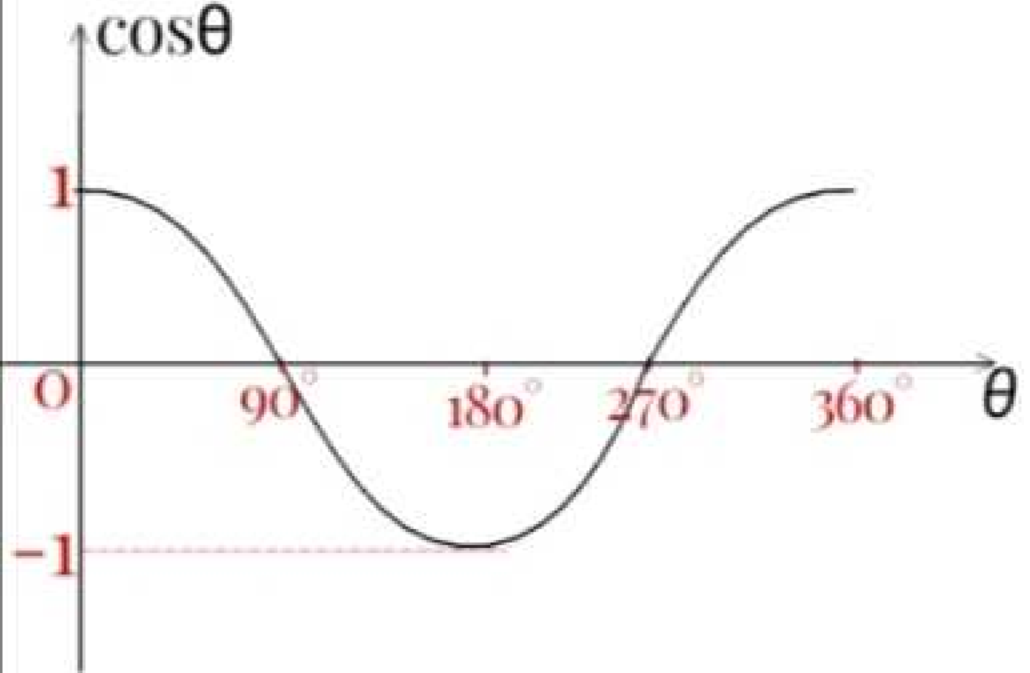
(ii) $\frac{2}{5}\pi^c$

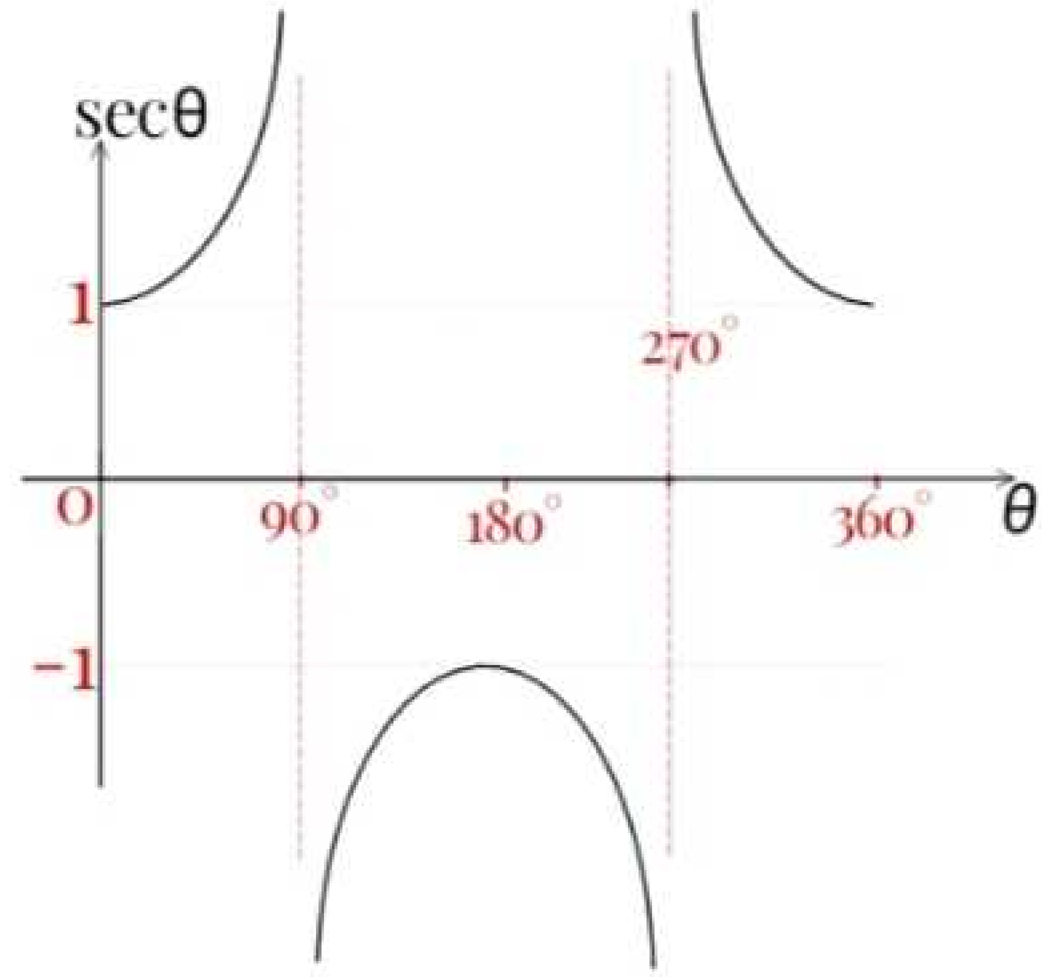
(iii) $\frac{2}{3}\pi^c$

leaving the answers in exact form

Graphs

1. A graph of $y = \cos\theta$ vs $y = \sec\theta$ for $0^\circ \le \theta \le 360^\circ$

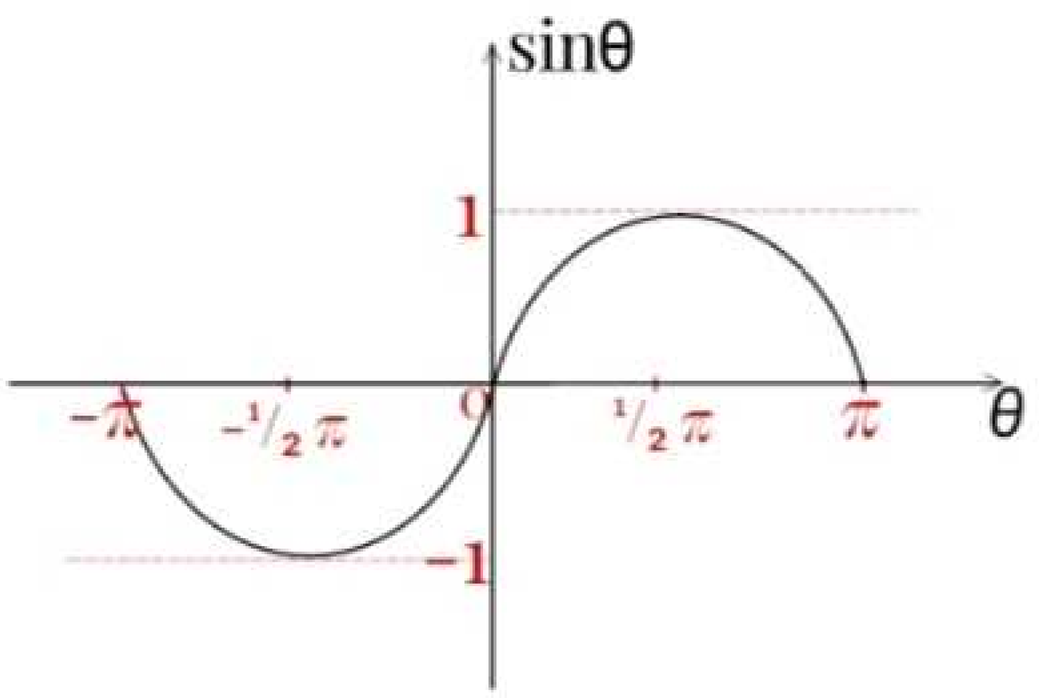


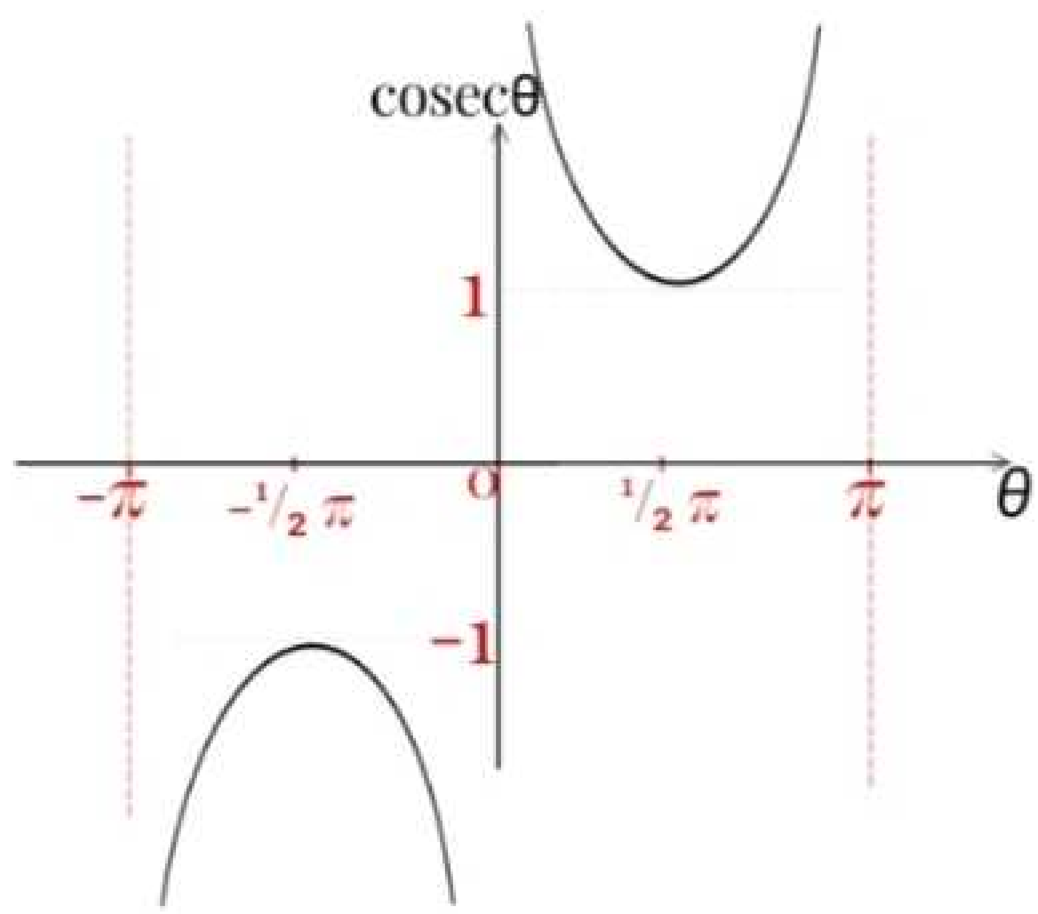


For a graph of $y = \sec\theta$:

- we've assymptotes at $\theta = 90^\circ$ and 270°.
- an assymptote is a value that can shouldn't be taken by a variable otherwise the function would become undefined.
- $\sec\theta = \dfrac{1}{\cos\theta}$ and so $\cos\theta$ shouldn't equals zero.

2. A graph of $y = \sin\theta$ vs $y = \operatorname{cosec}\theta$ for $-\pi \le \theta \le \pi$





- assymptotes at $\theta = -\pi, 0, \pi$
- that's where $\sin\theta$ is zero

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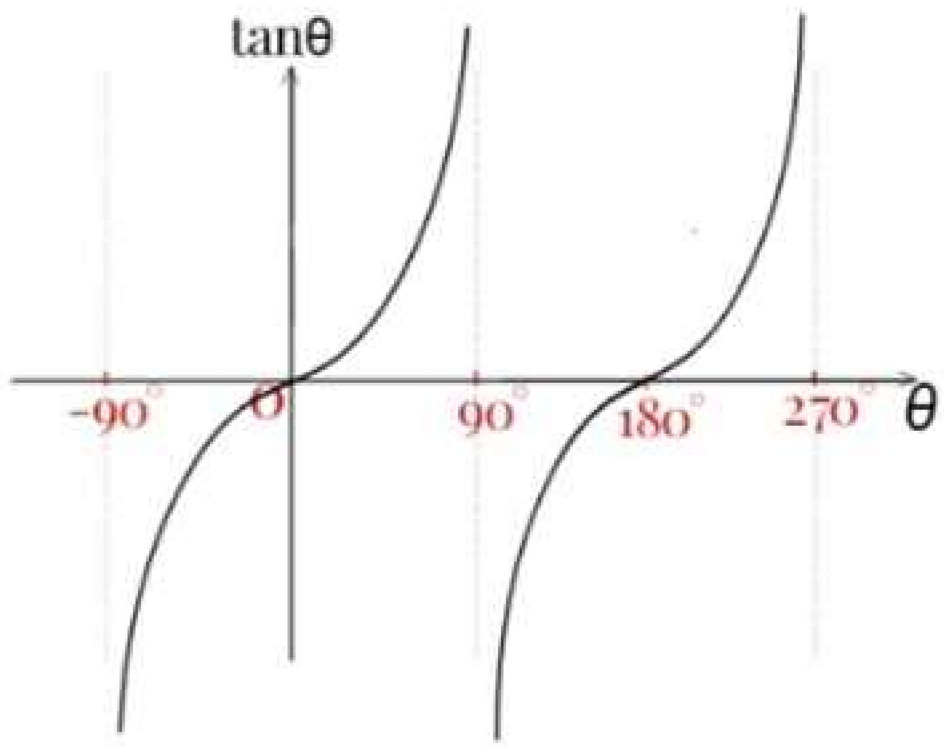
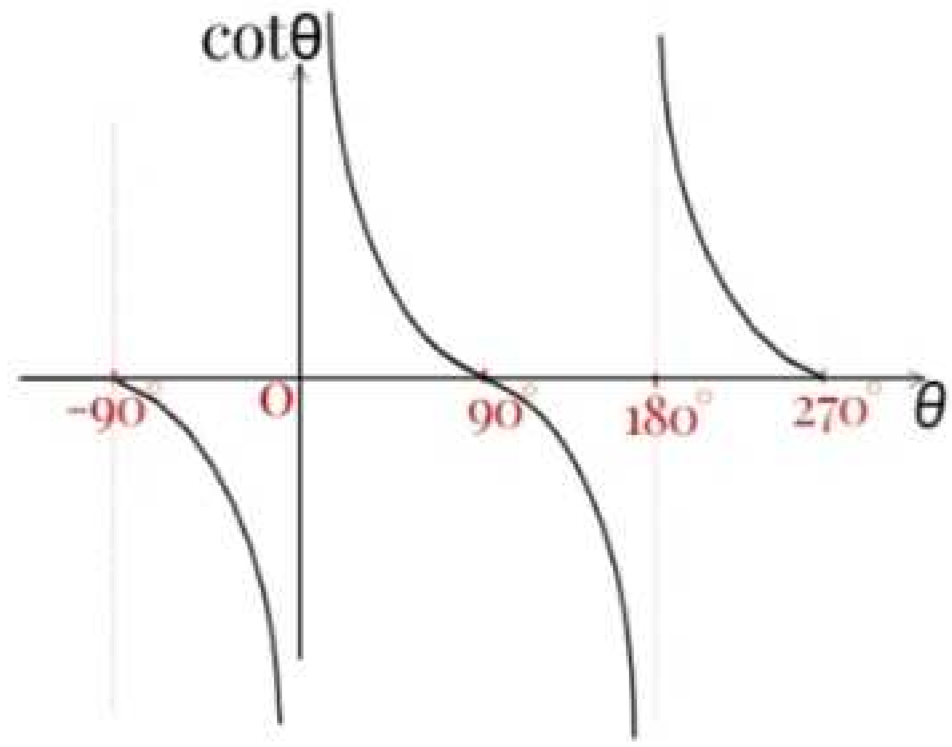
3. A graph of $y = \tan\theta$ vs $y = \cot\theta$ for $-90^\circ \leq \theta \leq 270^\circ$





$\tan\theta = \frac{\sin\theta}{\cos\theta}$ ∴ assymptotes are where $\cos\theta = 0$ $\{-90^\circ, 90^\circ, 270^\circ\}$

Practice

Sketch a graph of

(a)(i) $y = \cos\theta$ (ii) $y = \sec\theta$

(b)(i) $y = \sin\theta$ (ii) $y = \text{cosec}\theta$

(c)(i) $y = \tan\theta$ (ii) $y = \cot\theta$

for any range of values of θ.

Exact values of trigonometry functions

- Following is a table with exact values for some trig functions.
- These will one have to know by heart.

x^c	$\pi/6$	$\pi/4$	$\pi/3$
x°	30°	45°	60°
$\sin x$	$1/2$	$\sqrt{2}/2$ or $1/\sqrt{2}$	$\sqrt{3}/2$
$\cos x$	$\sqrt{3}/2$	$\sqrt{2}/2$ or $1/\sqrt{2}$	$1/2$
$\tan x$	$\sqrt{3}/3$ or $1/\sqrt{3}$	1	$\sqrt{3}$

- we can apply these in finding the the cosine, sine or tangent of an angle which can be expressed as sum/ diffrence of the angles in table without using a calculator.

Example

1) Without using a calculator, find the exact value for

(a) $\cos 105^\circ$

(b) $\sin 135^\circ$

(c) $\tan 15^\circ$

sol :

1(a) $\cos 105^\circ = \cos(60^\circ + 45^\circ)$ $\{\cos(A+B) = \cos A\cos B - \sin A\sin B\}$

$= \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ$

$= (1/2)(\sqrt{2}/2) - (\sqrt{3}/2)(\sqrt{2}/2)$

$= \sqrt{2}/4 - \sqrt{6}/4$

$= \boxed{(\sqrt{2} - \sqrt{6})/4}$

(b) $\sin 135^\circ = \sin(90^\circ + 45^\circ)$

$= \sin 90^\circ \cos 45^\circ + \sin 45^\circ \cos 90^\circ$

$= (\sqrt{2}/2)\sin 90^\circ + (\sqrt{2}/2)\cos 90^\circ$

But $\sin 90^\circ = \sin(45^\circ + 45^\circ)$

$= 2\sin 45^\circ \cos 45^\circ$

$= 2(\sqrt{2}/2)(\sqrt{2}/2) = 1$

$$\text{And } \cos 90^\circ = \cos(45^\circ + 45^\circ)$$
$$= \cos^2 45^\circ - \sin^2 45^\circ$$
$$= (\sqrt{2}/2)^2 - (\sqrt{2}/2)^2 = 0$$
$$\therefore \sin 135^\circ = (\sqrt{2}/2)(1) + (\sqrt{2}/2)(0)$$
$$\boxed{= \sqrt{2}/2}$$

$$\text{(c) } \tan 15^\circ = \tan(45^\circ - 30^\circ)$$
$$= \frac{\tan 45^\circ - \tan 30^\circ}{1 + \tan 45^\circ \tan 30^\circ}$$
$$= \frac{1 - \sqrt{3}/3}{1 + (1)(\sqrt{3}/3)}$$
$$= \frac{\left(\frac{3-\sqrt{3}}{3}\right)}{\left(\frac{3+\sqrt{3}}{3}\right)}$$
$$= \frac{3-\sqrt{3}}{3} \times \frac{3}{3+\sqrt{3}}$$
$$= \frac{3-\sqrt{3}}{3+\sqrt{3}}$$
$$= \frac{(3-\sqrt{3})(3-\sqrt{3})}{(3+\sqrt{3})(3-\sqrt{3})}$$
$$= \frac{3^2 - 6\sqrt{3} + 3}{9-3}$$
$$= \frac{12 - 6\sqrt{3}}{6} \quad \boxed{= 2 - \sqrt{3}}$$

Practice

Find the exact values of each of the following without using a calculator.

(a) $\cos 75^\circ$

(b) $\sin 150^\circ$

(c) tan 105°

Ans :

(a) $-\frac{2+\sqrt{6}}{4}$ (b) $\frac{1}{2}$ (c) $-2-\sqrt{3}$

Solving trigonometric equations

- First obtain the principal value(s) by solving the equation and then use the graphical method to find the other roots within the specified range.

Examples

1. Find all values of θ for which $0° < \theta < 360°$ that satisfy the equation $\sin(^1/_2\,\theta) = {}^1/_2$

sol :

$\sin(^1/_2\,\theta) = {}^1/_2$, let $^1/_2\,\theta = u \Rightarrow \theta = 2u$

$\sin u = {}^1/_2$

$\therefore u = \sin^{-1}(^1/_2)$

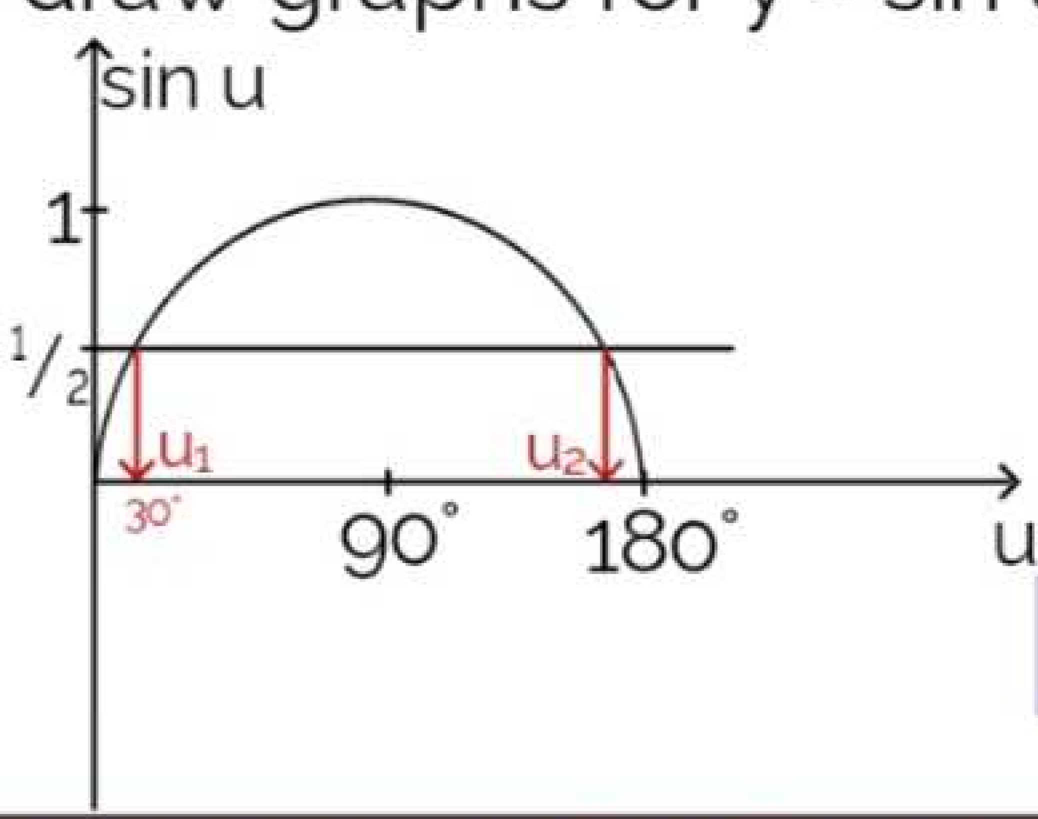
$= 30°$ } P.V

Range : $0° < \theta < 360°$

$\therefore\ 0° < 2u < 360°$

$\Rightarrow\ 0° < u < 180°$

Now draw graphs for $y = \sin u$ and $y = {}^1/_2$ for $0° < u < 180°$



$u_1 = 30°, \therefore \theta_1 = 2(30°) = 60°$

$u_2 = 180° - 30° = 150°, \therefore \theta_2 = 2(150°)$
$= 300°$

$\boxed{\therefore \theta = 60°, 300°}$

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2. Solve the equation $2\cos^2\theta + \cos\theta - 1 = 0$ for $0^\circ < \theta < 360^\circ$

sol:

$2\cos^2\theta + \cos\theta - 1 = 0$, let $\cos\theta = u$

$2u^2 + u - 1 = 0$

Now solve the quadratic equation :

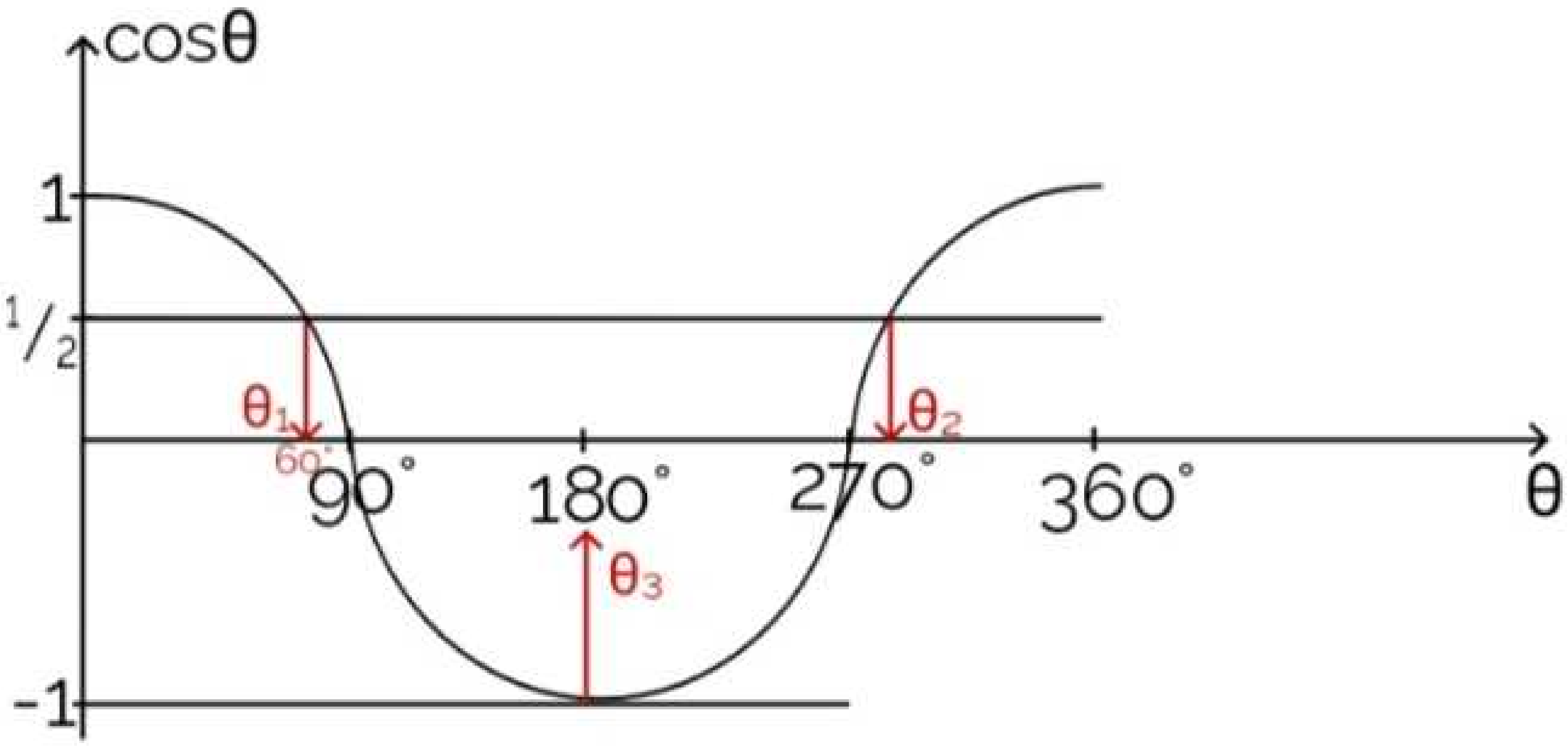
$u = -1$ OR $u = {}^1/_2$

$\therefore \cos\theta = -1$ OR $\cos\theta = {}^1/_2$

$\theta = \cos^{-1}(-1)$ OR $\theta = \cos^{-1}({}^1/_2)$

$= 180^\circ$ OR $= 60^\circ$ } P.Vs

Now draw graphs for $y = \cos\theta$, $y = -1$ and $y = {}^1/_2$



$\therefore \theta_1 = 60^\circ$

$\theta_2 = 360^\circ - 60^\circ$

$= 300^\circ$

$\theta_3 = 180^\circ$

$$\boxed{\therefore \theta = 60^\circ, 300^\circ, 180^\circ}$$

3. Solve the equation $2\sin\theta = \sec\theta$, for $0^\circ < \theta < 360^\circ$

sol:

$2\sin\theta = \sec\theta$

$2\sin\theta = \dfrac{1}{\cos\theta}$

$2\sin\theta\cos\theta = 1$

$\sin 2\theta = 1$ let $2\theta = u \Rightarrow \theta = \dfrac{u}{2}$

$\therefore \sin u = 1$

$u = \sin^{-1}(1)$

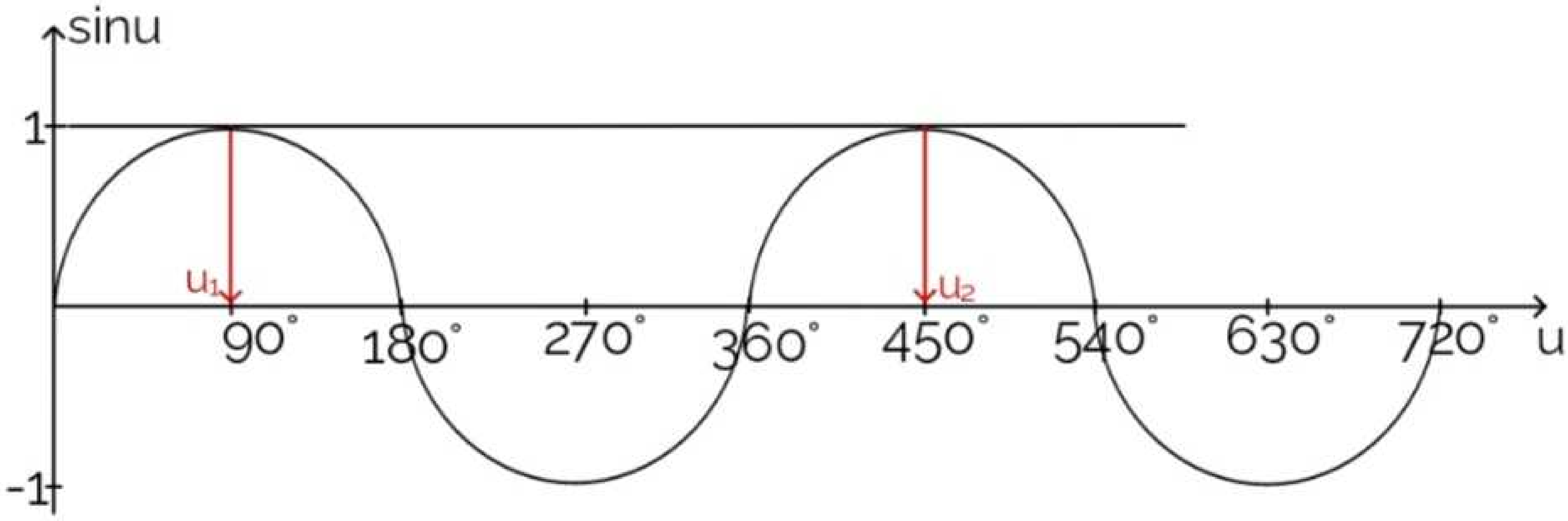
$= 90^\circ$

$0^\circ < \frac{u}{2} < 360^\circ$

$2(0^\circ) < u < 2(360^\circ)$

$0^\circ < u < 720^\circ$

Now draw graphs for $y = \sin u$ and $y = 1$ for $0^\circ < u < 720^\circ$



$u_1 = 90^\circ, \therefore \theta_1 = \frac{90}{2} = 45^\circ$

$u_2 = 450^\circ, \therefore \theta_2 = \frac{450}{2} = 225^\circ$

$\therefore \theta = 45^\circ, 225^\circ$

Practice

1) Solve, for $0^\circ < \theta < 360^\circ$, the equation $3\sec\theta + \tan^2\theta + 2 = 0$.
2) Solve, for $0^\circ < \theta < 360^\circ$, the equation $\cos(\theta + 15) = \frac{\sqrt{3}}{2}$
3) Solve, for $0^\circ < \theta < 360^\circ$, the equation $2\cos^2\theta + \sin\theta = 1$

Past Exam Questions

ZIMSEC P1 Nov 2013

12(a) Prove the identity

$$\frac{1 + \sin 2\theta}{1 - \sin 2\theta} \equiv \frac{(\tan\theta + 1)^2}{(\tan\theta - 1)^2}$$ [4]

(b) Solve the equation

$\tan\theta\cos 2\theta = \mathrm{Sin}\theta$ for $0^\circ \leq \theta \leq 360$ [6]

sol :

12a) LHS : $\dfrac{1 + \sin 2\theta}{1 - \sin 2\theta}$

$$= \frac{\cos^2\theta + \sin^2\theta + 2\cos\theta\sin\theta}{\cos^2\theta + \sin^2\theta - 2\cos\theta\sin\theta}$$

$$= \frac{\sin^2\theta + 2\cos\theta\sin\theta + \cos^2\theta}{\cos^2\theta - 2\cos\theta\sin\theta + \sin^2\theta}$$

But $(a + b)^2 <=> a^2 + 2ab + c^2$

OR

$(a - b)^2 <=> a^2 - 2ab + c^2$

$$= \frac{(\sin\theta + \cos\theta)^2}{(\sin\theta - \cos\theta)^2}$$

$$= \frac{\left(\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\cos\theta}\right)^2}{\left(\frac{\sin\theta}{\cos\theta} - \frac{\cos\theta}{\cos\theta}\right)^2}$$

Divide every term by $\cos\theta$

$$= \frac{(\tan\theta + 1)^2}{(\tan\theta - 1)^2}$$

$\equiv$ RHS

(b) $\tan\theta\cos 2\theta = \sin\theta$, for $0^\circ \leq \theta \leq 360^\circ$

$$\frac{\sin\theta}{\cos\theta} . \cos 2\theta = \sin\theta$$

$$\frac{\cos 2\theta}{\cos\theta} = 1$$

$\cos 2\theta = \cos\theta$

$\cos^2\theta - \sin^2\theta = \cos\theta$

$\cos^2\theta - (1 - \cos^2\theta) - \cos\theta = 0$

$\cos^2\theta - 1 + \cos^2\theta - \cos\theta = 0$

$2\cos^2\theta - \cos\theta - 1 = 0$ let $\cos\theta = u$

$2u^2 - u - 1 = 0$

solve the quadratic equation :

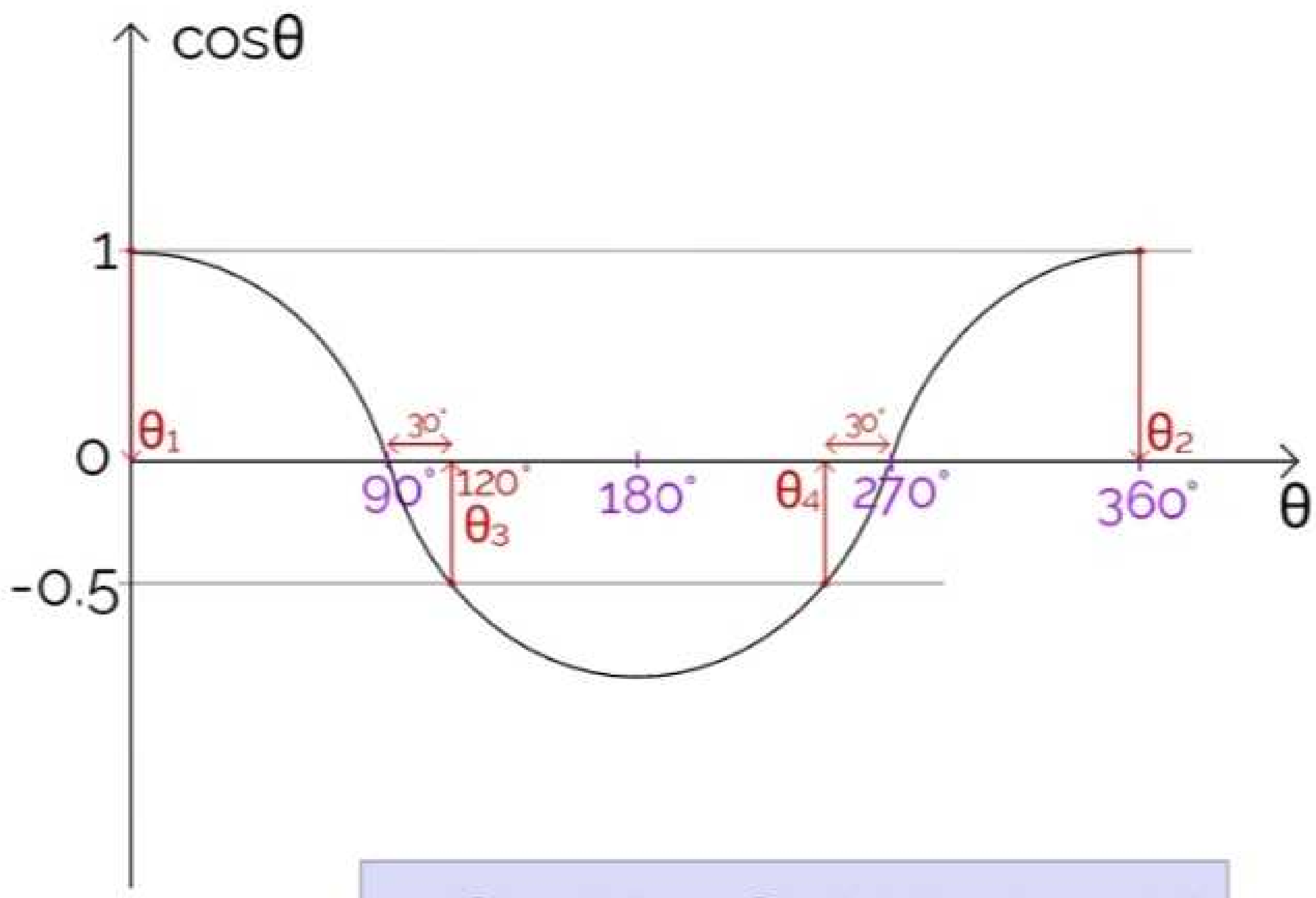
$u = -0.5$ OR $u = 1$

$\cos\theta = -0.5$ OR $\cos\theta = 1$

$\therefore \theta = \cos^{-1}(-0.5)$ OR $\theta = \cos^{-1}(1)$

$= 120^\circ$ OR $= 0^\circ$ } P.Vs

Now draw the graphs for $y = \cos\theta$, $y = 0.5$ and $y = 1$ for $0^\circ \le \theta \le 360^\circ$



$\therefore \theta_1 = 0^\circ$

$\theta_2 = 360^\circ$

$\theta_3 = 120^\circ$

$\theta_4 = 270^\circ - 30^\circ = 240^\circ$

$\therefore \theta = 0^\circ, 360^\circ, 120^\circ, 240^\circ$

ZIMSEC P2 Nov 2021

2(a) Prove the identity

$$\text{cosec}2\theta + \cot 2\theta \equiv \cot\theta \qquad [3]$$

(b) Hence or otherwise solve the equation

$\text{cosec}2\theta + \cot 2\theta = 0.5$, for $0 \le \theta \le 360$, giving the answer to 2d.p. [4]

sol :

2(a) LHS : $\text{cosec}2\theta + \cot 2\theta$

$$= \frac{1}{\sin 2\theta} + \frac{\cos 2\theta}{\sin 2\theta}$$

$$= \frac{1 + \cos^2\theta - \sin^2\theta}{\sin 2\theta}$$

[$\cos 2\theta = \cos^2\theta - \sin^2\theta$, $\sin 2\theta = 2\cos\theta\sin\theta$]

$$= \frac{\cos^2\theta + \sin^2\theta + \cos^2\theta - \sin^2\theta}{2\cos\theta\sin\theta}$$

[$\cos^2\theta + \sin^2\theta = 1$]

$$= \frac{\cancel{2}\cos^{\cancel{2}}\theta}{\cancel{2}\cancel{\cos\theta}\sin\theta}$$

$$= \frac{\cos\theta}{\sin\theta}$$

$$= \cot\theta \equiv \text{RHS}$$

(b) $\text{cosec}2\theta + \cot 2\theta = 0.5$, for $0 \le \theta \le 360$

But $\text{cosec}2\theta + \cot 2\theta = \cot\theta$

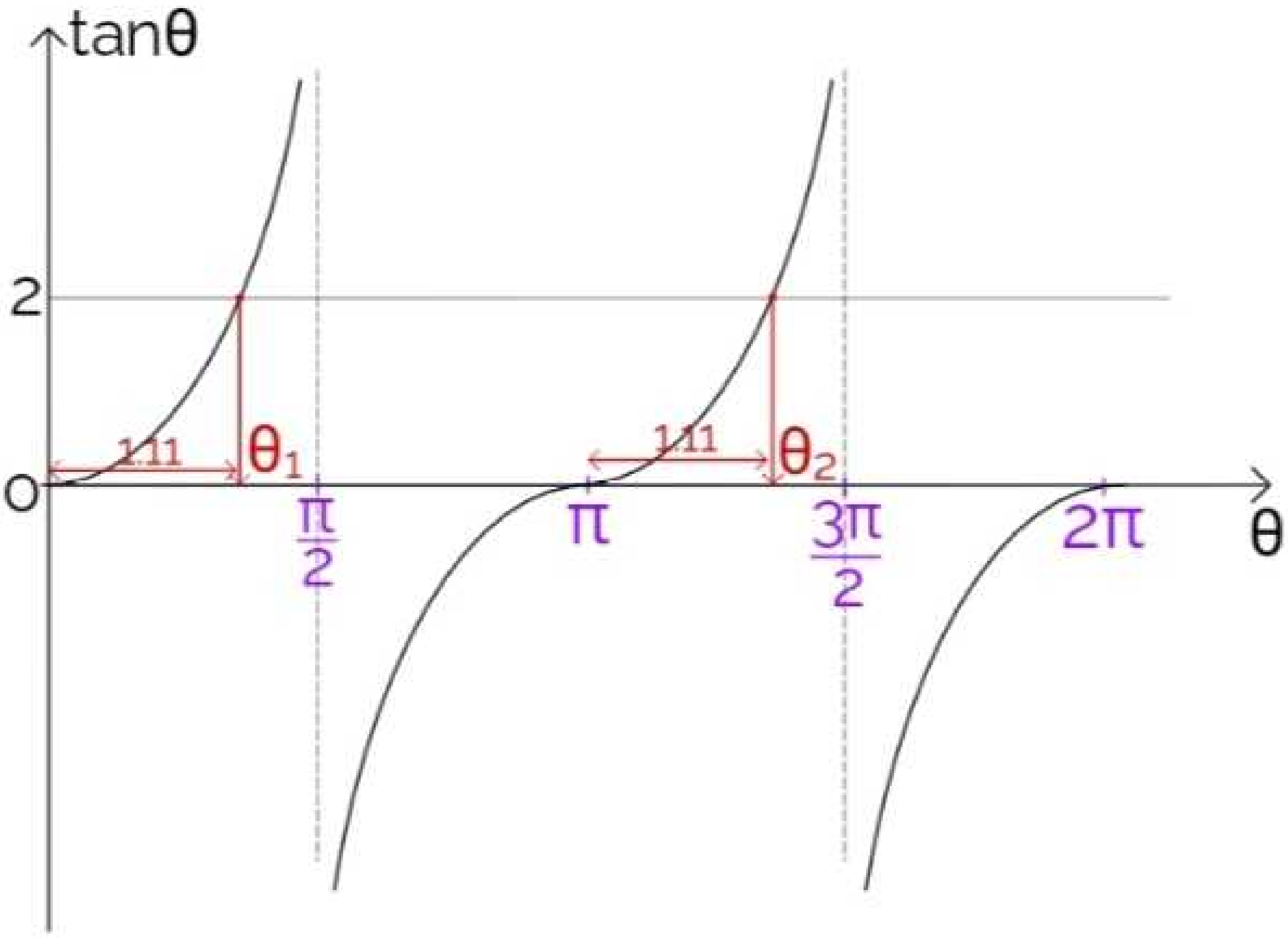
$\therefore \quad \cot\theta = 0.5$

$$\frac{1}{\tan\theta} = 0.5$$

$$2 = \tan\theta$$

$\theta = \tan^{-1}(2) = 1.11$ } P.V

Now draw graphs for $y = \tan\theta$ and $y = 2$, for $0 \leq \theta \leq 2\pi$



$$\therefore \theta_1 = 1.11$$
$$\theta_2 = \pi + 1.11$$
$$= 4.25$$

$$\therefore \theta = 1.11,\ 4.25$$

<u>Expressing $a\cos\theta + b\sin\theta$ in the form $R\cos(\theta - \alpha)$</u>

- we use the following formular :

$$a\cos\theta \pm b\sin\theta = R\cos(\theta \pm \alpha)$$

reverse the sign

$\to a\cos\theta + b\sin\theta = R\cos(\theta - \alpha)$

$\to a\cos\theta - b\sin\theta = R\cos(\theta + \alpha)$

where :

$R = \sqrt{a^2 + b^2}$ and

$\alpha = \tan^{-1}(\frac{b}{a})$

Examples

1. Express $5\cos\theta - 12\sin\theta$ in the form $R\cos(\theta + \alpha)$, where $R > 0$ and $0 \le \alpha \le 360$, stating the value of R and giving the value of α correct to 0.1°.

sol :

$5\cos\theta - 12\sin\theta = R\cos(\theta + \alpha)$,

But $R = \sqrt{5^2 + 12^2}$

$= 13$

And $\alpha = \tan^{-1}(\frac{12}{5})$

$= 67.4^\circ$

$\therefore 5\cos\theta - 12\sin\theta = 13\cos(\theta + 67.4^\circ)$

2. Solve the equation $\sin x + \cos x = 1$ for $0^\circ \le x \le 180^\circ$. giving your answer correct to 1d.p

sol :

$\sin x + \cos x = 1$ But

$\sin x + \cos x = R\cos(x - \alpha)$, where :

$R = \sqrt{1^2 + 1^2}$ and $\alpha = \tan^{-1}(\frac{1}{1})$

$= \sqrt{2}$ $= 45^\circ$

$\therefore \sqrt{2}\cos(x - 45^\circ) = 1$ for $0^\circ \le x \le 180^\circ$

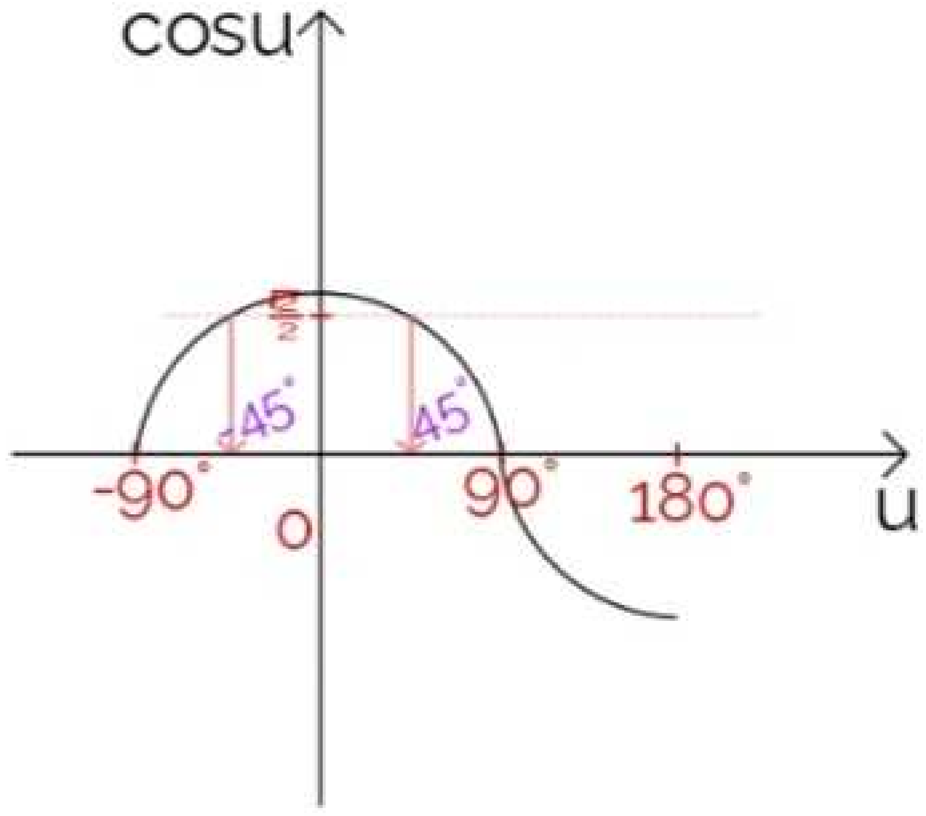
let $x - 45^\circ = u \Rightarrow x = u + 45^\circ$

$\therefore \sqrt{2}\cos u = 1$ for $0^\circ \le u + 45^\circ \le 180^\circ$

$u = \cos^{-1}(\frac{1}{\sqrt{2}})$ $\Rightarrow 0^\circ - 45^\circ \le u \le 180^\circ - 45^\circ$

$= 45^\circ$ } P.V $\Rightarrow -45^\circ \le u \le 135^\circ$

Now draw grahs for $y = \cos u$ and $y = \frac{1}{\sqrt{2}} / \frac{\sqrt{2}}{2}$ for $-90^\circ \le u \le 180^\circ$



Note : - It doesn't matter even if the graph exceeds the specified range a little bit, all you have to make sure is that your solutions are within the range.
- It's always a good practice to draw a graph with the limits that are a multiple of 90° and closest to specified range.

$\therefore u_1 = -45^\circ$
$u_2 = 45^\circ$

$\therefore x_1 = -45^\circ + 45^\circ = 0^\circ$
$x_2 = 45^\circ + 45^\circ = 90^\circ$

$\boxed{\therefore x = 0^\circ, 90^\circ}$

Practice

1. Given that $3\cos x - 4\sin x \equiv R\cos(x + \alpha)$, where $R > 0$ and $0^\circ \leq \alpha \leq 90^\circ$, find the values of R and α, giving the value of α correct to two decimal places. Hence solve the equation $3\cos 2\theta - 4\sin 2\theta = 1$, for $0^\circ \leq \theta \leq 360^\circ$, giving your answers correct to 1d.p.

Ans

$\theta = 6.7^\circ, 120^\circ, 186.7^\circ, 326.8^\circ$

Past Exam Question

ZIMSEC Nov 2003 P1

Express $\sqrt{6}\cos\theta - \sqrt{2}\sin\theta$ in the form $R\cos(\theta + \alpha)$, where $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$ [2]

Hence or otherwise find the solution of the equation
$(\sqrt{6}\cos\theta - \sqrt{2}\sin\theta)^2 = 4$ for $0 \le \theta \le 2\pi$ [4]

sol :

$\sqrt{6}\cos\theta - \sqrt{2}\sin\theta = R\cos(\theta + \alpha)$ where :

$R = \sqrt{(\sqrt{6})^2 + (\sqrt{2})^2}$ and $\alpha = \tan^{-1}(\frac{\sqrt{2}}{\sqrt{6}})$

$= \sqrt{8}$ $= \frac{\pi}{6}$

$\therefore \sqrt{6}\cos\theta - \sqrt{2}\sin\theta = \sqrt{8}\cos(\theta + \frac{\pi}{6})$

Equation :

$(\sqrt{6}\cos\theta - \sqrt{2}\sin\theta)^2 = 4$, for $0 \le \theta \le 2\pi$

$[\sqrt{8}\cos(\theta + \frac{\pi}{6})]^2 = 4$

$(\sqrt{8})^2\cos^2(\theta + \frac{\pi}{6}) = 4$

$8\cos^2(\theta + \frac{\pi}{6}) = 4$

$\cos^2(\theta + \frac{\pi}{6}) = \frac{1}{2}$ let $u = \theta + \frac{\pi}{6}$ $\Rightarrow \theta = u - \frac{\pi}{6}$

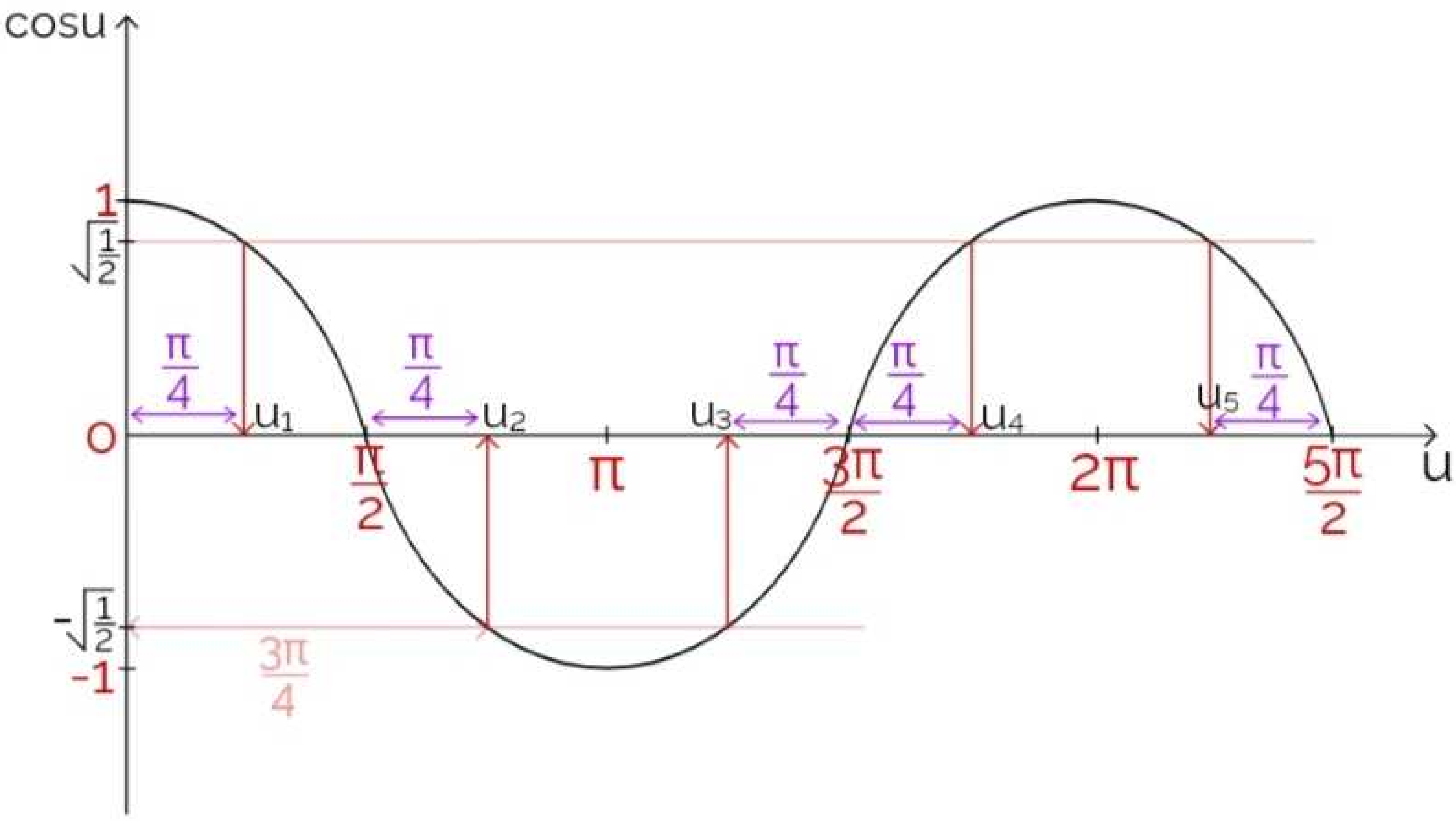
$\therefore \cos^2 u = \frac{1}{2}$ for $0 \le u - \frac{\pi}{6} \le 2\pi$

$\cos u = \pm\sqrt{\frac{1}{2}}$ $\Rightarrow 0 + \frac{\pi}{6} \le u \le 2\pi + \frac{\pi}{6}$

$u = \cos^{-1}(\sqrt{\frac{1}{2}})$ OR $\cos^{-1}(-\sqrt{\frac{1}{2}})$ $\Rightarrow \frac{\pi}{6} \le u \le \frac{13\pi}{6}$

$u = \frac{\pi}{4}$ OR $u = \frac{3\pi}{4}$ } P.Vs

Now draw graphs for $y = \cos u$, $y = \sqrt{\frac{1}{2}}$ and $y = -\sqrt{\frac{1}{2}}$ for $0 \le u \le \frac{5\pi}{2}$



$\therefore u_1 = \frac{\pi}{4}$ $\therefore \theta_1 = \frac{\pi}{4} - \frac{\pi}{6} = \frac{\pi}{12}$

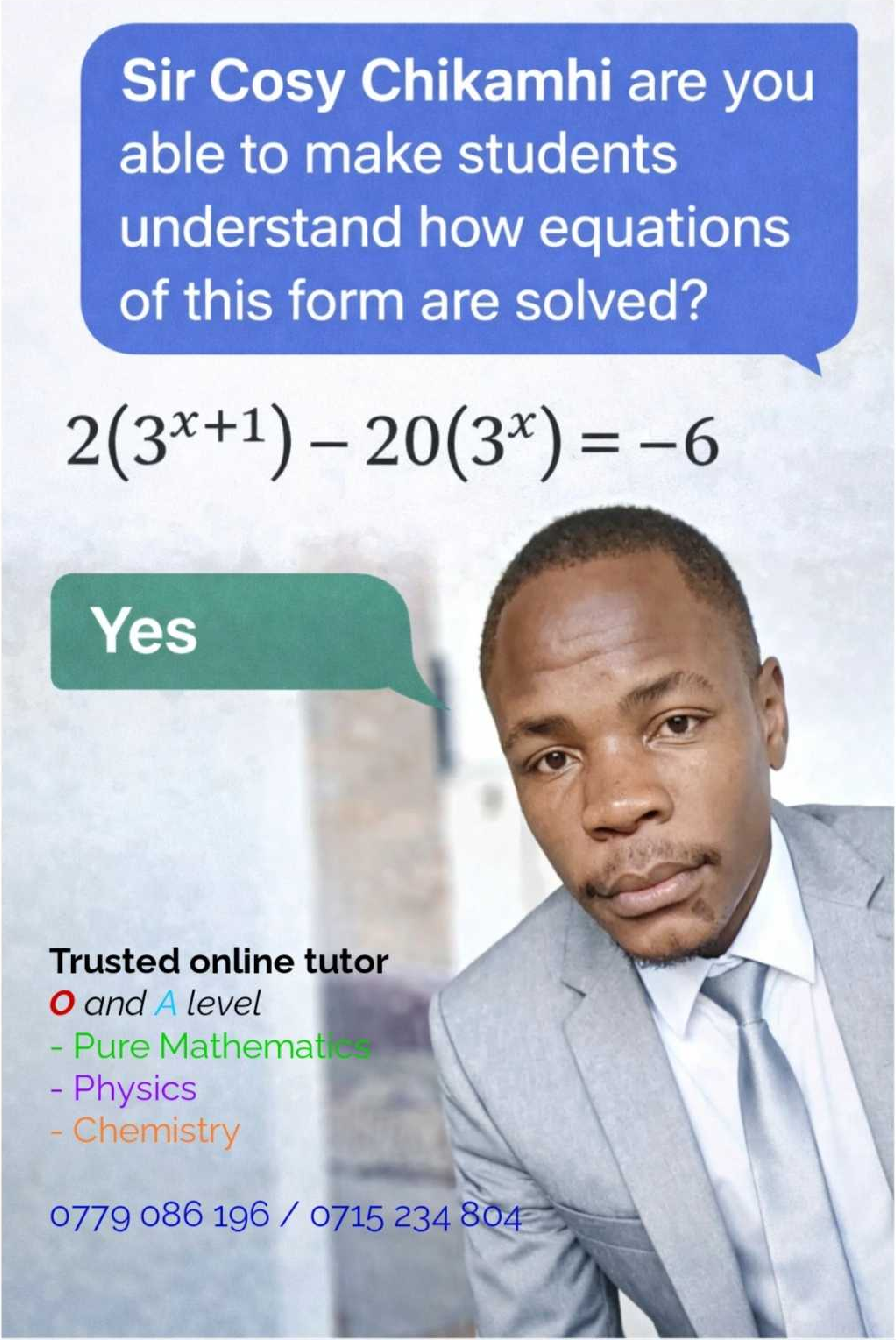
$u_2 = \frac{3\pi}{4}$ $\theta_2 = \frac{3\pi}{4} - \frac{\pi}{6} = \frac{7\pi}{12}$

$u_3 = \frac{3\pi}{2} - \frac{\pi}{4} = \frac{5\pi}{4}$ $\theta_3 = \frac{5\pi}{4} - \frac{\pi}{6} = \frac{13\pi}{12}$

$u_4 = \frac{3\pi}{2} + \frac{\pi}{4} = \frac{7\pi}{4}$ $\theta_4 = \frac{7\pi}{4} - \frac{\pi}{6} = \frac{19\pi}{12}$

$u_5 = \frac{5\pi}{2} - \frac{\pi}{4} = \frac{9\pi}{4}$ out of range, $\frac{9\pi}{4} > \frac{13\pi}{4}$

$$\boxed{\therefore \theta = \frac{\pi}{12}, \frac{7\pi}{12}, \frac{13\pi}{12}, \frac{19\pi}{12}}$$


Sir Cosy Chikamhi are you able to make students understand how equations of this form are solved?
$2(3^{x+1}) - 20(3^{x}) = -6$
Yes
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1John3:1

Behold, what manner of love the Father hath bestowed upon us, that we should be called the sons of God: therefore the world knoweth us not, because it knew him not.

John1:12

But as many as received him, to them gave he power to become the sons of God, even to them that believe on his name:

John3:16

For God so loved the world, that he gave his only begotten Son, that whosoever believeth in him should not perish, but have everlasting life.